



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

27-06-2021

JEE(Main + Advance) : Nurture Course (PHASE : S (TEPS))

Test Type : REVIEW TEST

Test Pattern : JEE (Advanced)

TEST DATE : 27-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	B,C	A,C	C	C,D	A,B,C,D	A,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	D	C	C	B	
SEC-3	Q.	1	2	3	4	5		
	A.	7	5	6	9	6		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	A,B,C,D	A,D	A,B	A,B,D	A,C,D	A,B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	A	B	D	B	A	D	
SEC-3	Q.	1	2	3	4	5		
	A.	9	2	8	3	3		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C,D	A,B	A,B,C,D	A,B,C	B,C,D	B,C	B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	D	A	C	D	
SEC-3	Q.	1	2	3	4	5		
	A.	3	6	5	7	8		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (B,C)

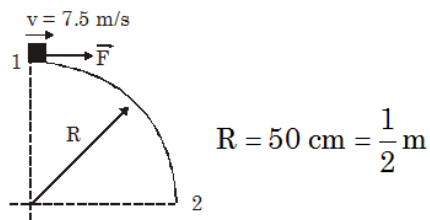
Viscosity of air increases with increase in temperature & viscosity of liquid decreases with increase in temperature

$$\text{and } |\eta_t| = \frac{2r^2(\rho_b - \rho_\ell)g}{9 \times \eta}$$

So in air η_1 decreases

and in liquid η_2 increases

2. Ans (B,C)

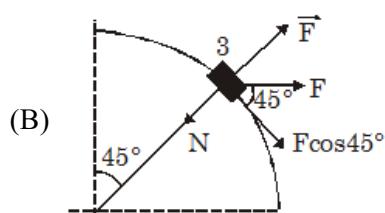


(A) From 1 to 2 applying work

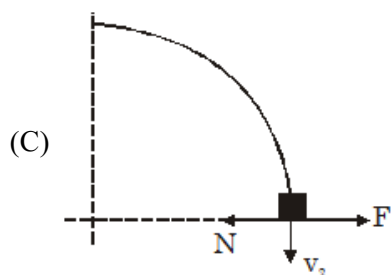
$$F \times R = \frac{1}{2}m(v_2^2 - v_1^2)$$

$$30 \times \frac{1}{2} = \frac{1}{2}(0.15)(v_2^2 - 7.5^2)$$

$$200 + \frac{225}{4} = v_2^2 = \sqrt{\frac{1025}{4}}$$

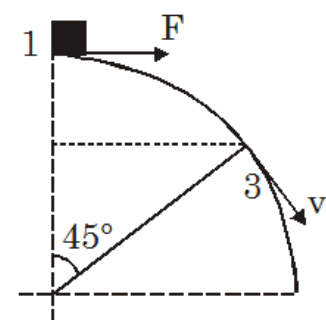


$$a_t = \frac{F \cos 45^\circ}{m} = 100\sqrt{2}$$



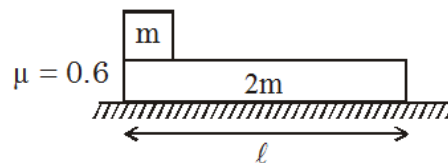
$$N - F = \frac{mv_2^2}{R}; N = F + \frac{mv_2^2}{R} \simeq 77N$$

(D) From part 1 to 3 WET



$$F \frac{R}{\sqrt{2}} = \frac{1}{2}m(v_3^2 - v_1^2); a_n = \frac{v_3^2}{R}$$

3. Ans (A,C)



(A) For maximum $\Delta T \rightarrow$ maximum heat $\rightarrow$ maximum work against friction

$$= \mu mg\ell = (3m)s \Delta T$$

$$\Rightarrow \Delta T = \frac{\mu g\ell}{3s} = \frac{0.6 \times 10 \times 10}{3 \times 2 \times 10^3}$$

(B) If block stops at mid point of plank:

$$\frac{\mu mg\ell}{2} = 3ms \Delta T; \Delta T \Rightarrow 0.5 \times 10^{-2} ^\circ C$$

(C) When block is projected with v_0 , $w_f = \Delta T$

$$-\mu mg\ell = \frac{1}{2}3mv_f^2 - \frac{1}{2}mv_0^2$$

$$\text{and } mv_0 = 3mv_f \Rightarrow v_f = \frac{v_0}{3}$$

$$-\mu mg\ell = \frac{1}{2}3m \left(\frac{v_0^2}{9} \right) - \frac{1}{2}mv_0^2$$

$$; -\mu mg\ell = -\frac{1}{2}mv_0^2; v_0^2 = 3\mu g\ell$$

Now when block is projected with $\frac{v_0}{2}$

$$m \frac{v_0}{2} = 3mv_f \Rightarrow v_f = \frac{v_0}{6}$$

$$w_f = \frac{1}{2}3m \left(\frac{v_0^2}{36} \right) - \frac{1}{2}m \frac{v_0^2}{4} = -\frac{1}{12}mv_0^2$$

$$\frac{1}{12}mv_0^2 = (3m)s\Delta T; \frac{m}{12}3\mu g\ell = 3ms\Delta T$$

$$\Delta T = \frac{\mu g\ell}{12s} = \frac{0.6 \times 10 \times 10}{12 \times 2 \times 10^3} = 2.5 \times 10^{-3} ^\circ C$$

4. Ans (C)

Rate of heat flow :

$$\frac{KA(400 - T)}{t} = \frac{d\theta}{dt} \quad \dots(i)$$

$$C = C_v + \frac{R}{1 - n} = \frac{3}{2}R + \frac{R}{1 + 1} = 2R$$

$$\frac{dQ}{dt} = nC \frac{dT}{dt} \quad \dots(ii)$$

From (i) & (ii)

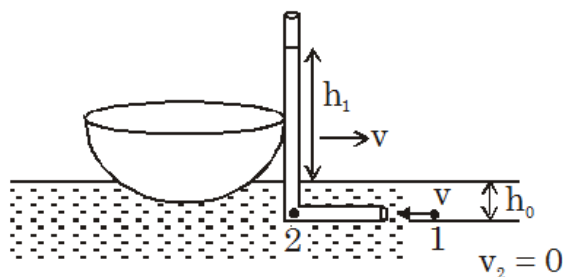
$$\frac{KA(400 - T)}{t} = nC \frac{dT}{dt}$$

$$KA \int_0^t \frac{dT}{dt} = nc \int_{300}^T \frac{dT}{400 - T}$$

$$\Rightarrow KA\ell nt = -nC \ln \left(\frac{400 - T}{100} \right)$$

5. Ans (C,D)

Applying bernoulli's theorem w.r.t boat in case-
(i) between position (i) & (ii)



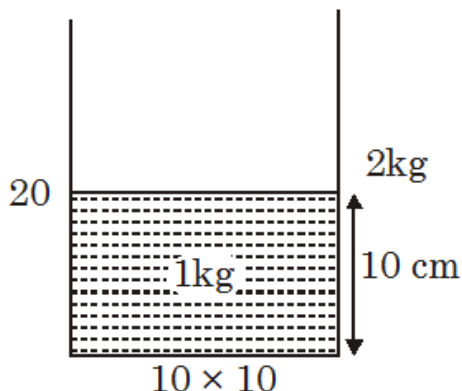
$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P_0 + \rho gh_0 + \frac{1}{2}\rho v^2 = P_0 + \rho gh_1 + \rho gh_0 + 0$$

$$h_1 = \frac{v^2}{2g}$$

similarly in (ii), (iii) & (iv) bernoulli equation gives same height.

6. Ans (A,B,C,D)



Volume of water = 1000 cm³

F to slide = $\mu(3g) = 6N$

Angle of water surface with horizontal $\tan\theta = a/g$

where $a = \frac{F - 6}{3}$ ($F > 6N$)

7. Ans (A,C)

$$P_{\text{net}} = mv \frac{dv}{dt}$$

$$P_{\text{engine}} - \mu mgv = mv \frac{dv}{dt}$$

$$\frac{dv}{dt} = \frac{P_{\text{engine}}}{mv} - \mu g$$

$$\text{For } v_{\text{max}} \Rightarrow \frac{dv}{dt} = 0$$

$$\frac{P_{\text{engine}}}{mv_{\text{max}}} - \mu g = 0; v_{\text{max}} = \frac{P_{\text{engine}}}{\mu mg}$$

PHYSICS

SEC-2

1. Ans (B)

(I) (iii) (R)

Monoatomic gas

$$f = 3, P v^{1.5} = \text{constant}$$

$$C = \frac{3}{2}R + \frac{R}{1 - 1.5} = -\frac{R}{2}$$

$$P_f = 0.5 \times 10^5$$

$$P v^{3/2} = \text{constant} \Rightarrow P \left(\frac{nRT}{P} \right)^{3/2} = \text{constant}$$

$$P^{-1/2} T^{3/2} = \text{Constant}$$

$$T^3 \propto P, P \downarrow \Rightarrow T \uparrow, Q = n C \Delta T$$

$$C < 0 \text{ \& } \Delta T < 0 \therefore Q > 0$$

2. Ans (C)

(III) (iii) (Q)

$$f_{\text{max}} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{\frac{1}{2}3 + \frac{1}{2}5}{1} = 4$$

$$P v^{1.5} = \text{constant}; T^3 \propto P$$

$$P_f = 2 \times 10^5 \Rightarrow P \uparrow \text{ so } T \uparrow$$

$$C = \frac{f}{2}R + \frac{R}{1 - n} = \frac{4}{2}R + \frac{R}{1 - 3/2} = 0$$

$$\therefore Q = 0$$

3. Ans (D)

(IV) (iv) (S)

$$P v^{-1} = \text{constant}$$

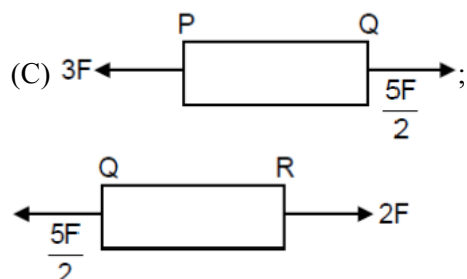
$$\frac{P \cdot P}{nRT} = \text{constant}$$

$$P^2 \propto T$$

$$P_F = 4 \times 10^{-5} \Rightarrow P \uparrow \Rightarrow T \uparrow$$

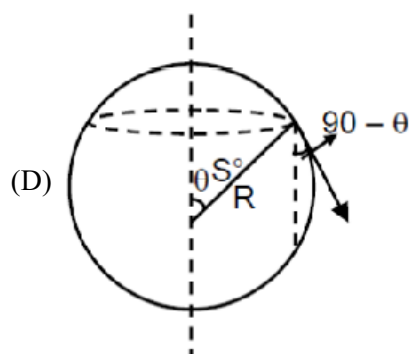
6. Ans (B)

For Q.4, 5 & 6



$$\Delta \ell_{PQ} = \frac{3F \ell / 2}{2A\sigma} + \frac{\frac{5F}{2} \ell}{2A\sigma} = \frac{3F \ell}{4A\sigma} + \frac{5F \ell}{8A\sigma} = \frac{11F \ell}{8A\sigma}$$

$$\Delta \ell_{QR} = \frac{2F \ell / 2}{2A\sigma} + \frac{\frac{5F}{2} \ell}{2A\sigma} = \frac{F \ell}{2A\sigma} + \frac{5F \ell}{8A\sigma} = \frac{9F \ell}{8A\sigma}$$



$$F = 4\pi(R \sin \theta) S \sin \theta = 4\pi R S \sin^2 \theta$$

$$F_{\text{net}} = 4\pi R S [\sin^2 53^\circ - \sin^2 37^\circ]$$

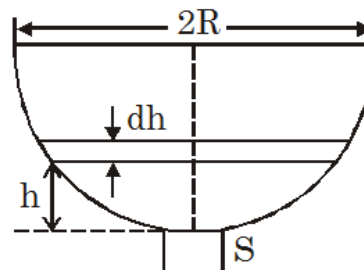
$$= 4\pi R S \left[\frac{16}{25} - \frac{9}{25} \right] = \frac{28\pi R S}{25}$$

$$\alpha = 28; \frac{3\alpha}{14} = 6; \beta = 25 \Rightarrow \frac{\beta}{5} = 5$$

PHYSICS

SEC-3

1. Ans (7)



Let the radius of the hemispherical vessel be R

and the area of orifice opening at S = a.

Consider an elemental volume of thickness dh

at height h from the bottom of the vessel. Now

$$\text{the radius } x^2 = R^2 - (R - h)^2 = 2Rh - h^2$$

The differential equation for rate of flow and

time is

$$dq = \pi x^2 dh = a \sqrt{2gh} dt;$$

$$dt = - \frac{\pi (2Rh - h^2) dh}{a \sqrt{2gh}}$$

$$\text{Integrating, } t = \frac{\pi}{a \sqrt{2g}} \int_{h_2}^{h_1} \left(2Rh^{\frac{1}{2}} - h^{\frac{3}{2}} \right) dh$$

where h_1 = height up to brim, h_2 = height up to

orifice

$$t = \frac{\pi}{a \sqrt{2g}} \left[\frac{4}{3} R h^{3/2} - \frac{2}{5} h^{5/2} \right]_0^R$$

$$= \frac{2\pi}{a \sqrt{2g}} \left[\frac{2}{3} R^{5/2} - \frac{1}{5} R^{5/2} \right] = \frac{14\pi R^{5/2}}{15a \sqrt{2g}}$$

2. Ans (5)

Let ℓ be the natural length of the rope and x the extension when the man hangs at one end.

$$\text{Then } W = n W \cdot \frac{x}{\ell} \text{ so } x = \frac{\ell}{n}$$

Hence total depth of man below the fixed point

$$= \ell + \frac{\ell}{n} = \frac{n+1}{n} \cdot \ell.$$

Now work done against gravity in going up a height

$$\frac{n+1}{n} \ell$$

$$\text{when the string is inextensible} = \frac{n+1}{n} \ell W \dots (1)$$

When the string is elastic the work done by the man

is less than the above by the amount of work done

by elasticity. i.e., by the amount of work done by

elastic string from tension W to tension zero,

$$\text{i.e., by} = \frac{1}{2}(W + 0) \times \frac{\ell}{n} = \frac{\ell W}{2n} \dots (2)$$

Work done when the string is elastic = (1) - (2)

$$= \frac{\ell W}{n} \left(n + 1 - \frac{1}{2} \right) = \frac{2n+1}{2n} \ell W \dots (3)$$

$$\text{Required ratio} = \frac{(3)}{(1)} = \frac{2n+1}{(2n+2)}$$

4. Ans (9)

The component of the plate's weight along the

plane. $F = 100 \cdot \sin 5^\circ$. Put the plate velocity to

V , oil film thickness to h , area of plate to A .

Use Newton's viscosity law.

$$\tau = \frac{F}{A} = \mu \cdot \frac{dv}{dy} = \mu \cdot \frac{V}{h}$$

$$\Leftrightarrow \frac{100 \cdot \sin 45^\circ}{1} = 0.1 \cdot \frac{V}{0.001}$$

$$\Rightarrow V = 8.7 \text{ cm/s}$$

5. Ans (6)

$$\left(P_0 + \frac{4s}{R} \right) \frac{4}{3} \pi R^3 = \left(\frac{P_0}{2} + \frac{16s}{5R} \right) \frac{4}{3} \pi \frac{125}{64} R^3$$

$$P_0 + \frac{4s}{R} = \left(\frac{P_0}{2} + \frac{16s}{5R} \right) \frac{125}{64}$$

$$P_0 + \frac{4s}{R} = \frac{125P_0}{128} + \frac{25s}{4R}$$

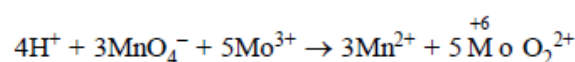
$$P_0 \left(\frac{128 - 125}{128} \right) = \frac{s}{R} \left(\frac{25 - 16}{4} \right)$$

$$P_0 = \frac{96S}{R} = \frac{16ns}{R} \Rightarrow n = 6$$

CHEMISTRY

SEC-1

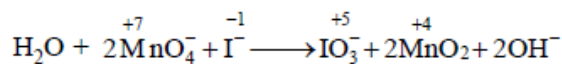
3. Ans (A,D)



$$20 \text{ ml} \quad n_f = 3$$

$$0.2M \quad 0.5 M$$

$$(nf = 5) \quad 10 \text{ ml}$$



$$0.1N$$

$$1 \text{ mmol} \quad 0.5 \text{ mmol}$$

$$(n_f = 3) \quad n_f = 6$$

$$V = 30 \text{ ml}$$

4. Ans (A,B)

Gas-A $\Rightarrow$ 200 per trillion (10^{12}) by volume

$\Rightarrow$ 200 L of gas A in 10^{12} L of solution

Using ideal gas equation ;

$$PV = nRT$$

$$\Rightarrow (1 \times 200) = n_A \times 0.08 \times 300$$

$$\therefore n_A = 8.33$$

$$\therefore [A]_{\text{gas}} =$$

$$\frac{\text{moles of gas A}}{\text{volume of solution (L)}} = \frac{8.33}{10^{12}} = 8.33 \times 10^{-12} \text{ M}$$

(A) $\rightarrow$ correct

Similarly,

For Gas-B $\Rightarrow$ 500 L of gas B in 10^{12} L of solution

$$\therefore n_B = \frac{PV}{RT}$$

$$n_B = \frac{0.5 \times 500}{0.08 \times 300} = 10.417 \text{ moles}$$

$$\therefore [B] = \frac{10.417}{10^{12}} = 1.04 \times 10^{-11} \text{ M (B)} \rightarrow \text{Correct}$$

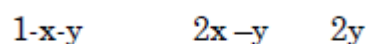
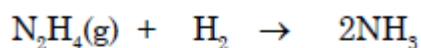
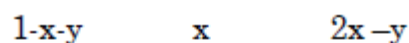
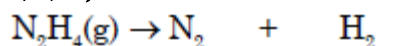
C) 27°C & 0.5 atm ; (Gas-A)

$$n_A = \frac{0.5 \times 200}{0.08 \times 300} = 4.167$$

$$\therefore [A] = \frac{4.1667}{10^{12}} = 4.167 \times 10^{-12} \text{ M}$$

(C) $\rightarrow$ Incorrect

5. Ans (A,B,D)



Total pressure

$$= 1 - x - y + x + 2x - y + 2y = 1.4$$

$$\Rightarrow 1 + 2x = 1.4$$

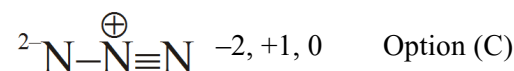
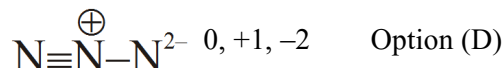
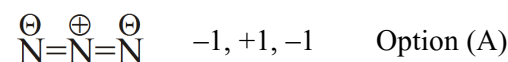
$$\Rightarrow x = 0.2$$

$$\Rightarrow P_{\text{H}_2} = 0.3 = 2x - y \Rightarrow y = 0.1 \text{ atm}$$

$$\Rightarrow K_{p1} = \frac{(0.2)(0.3)^2}{0.7} = \frac{18}{7} \times 10^{-2} (\text{atm})^2$$

$$\Rightarrow K_{p2} = \frac{(0.2)^2}{0.7 \times 0.3} = \frac{4}{21}$$

6. Ans (A,C,D)



(A,C,D) are correct option.

CHEMISTRY

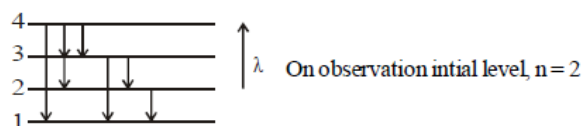
SEC-3

2. Ans (2)

Let initial state be 'n'.

6 different wavelengths $\Rightarrow$ final state = 4

3 photons have higher energy



3. Ans (8)

Initial situation

$$\text{Pair} + 70 \text{ cm Hg} = 76 \text{ cm Hg}$$

$$\Rightarrow \text{Pair} = 6 \text{ cm Hg}$$

Final situation :

Since tube is pushed downward, by 7 cm the column of air will decrease by 7. But due to increased air pressure, height of Hg column will also decrease. Let Hg column height decreased by x .

Length of air chamber

$$= 12 - (7 - x) = (5 + x)$$

$$\text{length of Hg} = (70 - x)$$

$$\Rightarrow P_{\text{air}} = 76 - (70 - x) = 6 + x$$

$$\Rightarrow \text{apply } P_1 V_1 = P_2 V_2$$

$$\Rightarrow 12 \times 6 = (5 + x)(6 + x)$$

$$\Rightarrow 72 = 30 + 11x + x^2$$

$$\Rightarrow x^2 + 11x - 42 = 0$$

$$\Rightarrow x = 3$$

4. Ans (3)



at $t = 0$ 1

$$t = t_{eq} 1 - \mu, \mu, \frac{\alpha}{2}$$

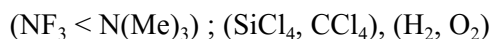
$$K_p = \frac{\left(\frac{\alpha}{1+\frac{\alpha}{2}} \cdot P\right)^2 \times \left(\frac{\frac{\alpha}{2}}{1+\frac{\alpha}{2}} \cdot P\right)}{\left(\frac{1-\frac{\alpha}{2}}{1+\frac{\alpha}{2}} \cdot P\right)^2}$$

$\alpha \Rightarrow$ small

$$K_p = \alpha^2 \times \frac{\alpha}{2} \times P$$

$$\Rightarrow \alpha = \left(\frac{2K_p}{P}\right)^{\frac{1}{3}}$$

5. Ans (3)



MATHEMATICS

SEC-1

1. Ans (A,B,C,D)

$$(2n+1)^2 - (2m+1)^2 = (2n+2m+2)(2n-2m) \\ = 4(n+m+1)(n-m)$$

$$\therefore (2n+1)^2 - (2m+1)^2 = 8k$$

$$\therefore a_4 - a_1 = 8d_1, a_5 - a_2 = 8d_2, a_6 - a_3 = 8d_3, \\ a_7 - a_4 = 8d_4, a_8 - a_5 = 8d_5, a_9 - a_6 = 8d_6$$

$$\therefore \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 - a_1 & a_5 - a_2 & a_6 - a_3 \\ a_7 - a_4 & a_8 - a_5 & a_9 - a_6 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ 8d_1 & 8d_2 & 8d_3 \\ 8d_4 & 8d_5 & 8d_6 \end{vmatrix} = 64 \begin{vmatrix} a_1 & a_2 & a_3 \\ d_1 & d_2 & d_3 \\ d_4 & d_5 & d_6 \end{vmatrix}$$

$\therefore$ the determinant is always a multiple of 4, 16, 64.

3. Ans (A,B,C,D)

$$P(x) = x^{32} - x^{25} + x^{18} - x^{11} + x^4 - x^3 + 1$$

$$\text{for } x < 0 \quad x^{32} > 0$$

$$-x^{25} > 0$$

$$x^{18} > 0$$

$$-x^{11} > 0$$

$$x^4 > 0$$

$$-x^3 > 0$$

$$\text{hence } P(x) > 0 \text{ for } x < 0 \quad \dots(i)$$

$$\text{for } x = 0; P(0) = 1 > 0 \quad \dots(ii)$$

$$\text{for } x > 1 \quad x^{32} - x^{25} > 0$$

$$x^{18} - x^{11} > 0$$

$$x^4 - x^{11} > 0$$

$$x^4 - x^3 > 0$$

$$1 > 0$$

$$\text{hence } P(x) > 0 \text{ for } x > 1 \quad \dots(iii)$$

Now for $x \in (0, 1)$

$$x^{32} > 0$$

$$x^{18} - x^{25} > 0$$

$$1 - x^3 > 0$$

$$\text{hence } P(x) > 0 \text{ for } x \in (0, 1) \quad \dots(iv)$$

hence from equation (i), (ii), (iii), (iv)

we can conclude that $P(x) > 0 \forall x \in \mathbb{R}$

hence no real root (option A)

So all roots are imaginary (option B)

$\therefore$ No real root present hence No. of negative roots are zero. (option C)

$$P(x) + P(-x) = 0$$

$$= (x^{32} + x^{18} + x^4 + 1) = 0$$

$$g(x) = x^{32} + x^{18} + x^4 + 1 = 0$$

$$g(x) > 0 \forall x \in \mathbb{R}$$

hence all roots are imaginary. (option D)

4. Ans (A,B,C)

$$S = 1 + \left(1 + \frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(1 + \frac{1}{2} + \frac{1}{2^2}\right) \left(\frac{1}{3^2}\right) + \dots \dots (1)$$

$$\frac{S}{3} = \frac{1}{3} + \left(1 + \frac{1}{2}\right) \left(\frac{1}{3^2}\right) + \dots \dots (2)$$

Equation (1) - (2)

$$\frac{2S}{3} = 1 + \frac{1}{6} + \frac{1}{6^2} + \dots \dots = \frac{6}{5}$$

$$S = \frac{18}{10} = \frac{9}{5}$$

5. Ans (B,C,D)

$$\tan A \tan B \tan C = \frac{3 + \sqrt{3}}{\sqrt{3} - 1} = 3 + 2\sqrt{3} =$$

$$\tan A + \tan B + \tan C$$

Using the identity,

$$\cos(A + B + C) = \cos A \cos B \cos C [1 - \Sigma \tan A \tan B]$$

$$\Rightarrow \Sigma \tan A \tan B = 5 + 4\sqrt{3}$$

Now,

$$x^3 - (3 + 2\sqrt{3})x^2 + (5 + 4\sqrt{3})x - (3 + 2\sqrt{3}) = 0$$

$$x = \tan A, \tan B, \tan C$$

It is not hard to observe that $x = 1$, is one of the root

$$\text{Hence } x = 1, \sqrt{3}, 2 + \sqrt{3}$$

6. Ans (B,C)

$$2 \cos^2 2x - 2 \cos 2x \cos \left(\frac{2014\pi^2}{x} \right) = -2 \sin^2 2x$$

$$\Rightarrow \cos 2x \cos \frac{2014\pi^2}{x} = 1$$

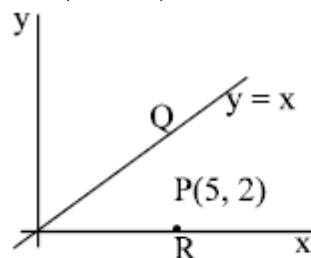
$$\cos 2x, \cos \left(\frac{2014\pi^2}{x} \right) \text{ should both be}$$

simultaneously 1 or -1.

positive solutions : $\pi, 19\pi, 53\pi, 1007\pi$

sum = 1080π .

7. Ans (B,C,D)



Take the image of P in $y = x$ to get P' as (2, 5)

Take the image of P in x axis to get P'' as (5, -2)

Write down the equation of line joining P' and P''

$$7x + 3y = 29$$

$$\text{solve with } y = x \text{ to get } Q \left(\frac{29}{10}, \frac{29}{10} \right)$$

$$\text{Solve with x - axis to get } R \left(\frac{29}{7}, 0 \right)$$

MATHEMATICS

SEC-2

1. Ans (B)

$$PA^2 = K^2 \cdot PB^2$$

$$\Rightarrow (x - 2)^2 + (y - 3)^2 = K^2(x - 5)^2 + K^2(y - 9)^2$$

$$\Rightarrow x^2(1 - K^2) + y^2(1 - K^2) - 2x(2 - 5K^2)$$

$$- 2y(3 - 9K^2) + (4^2 + 3^2 - 5^2K^2 - 9^2K^2) = 0$$

$$\Rightarrow x^2 + y^2 - 2x \left(\frac{5K^2 - 2}{K^2 - 1} \right) - 2y \left(\frac{9K^2 - 3}{K^2 - 1} \right) + \frac{(4^2 + 3^2 - 5^2K^2 - 9^2K^2)}{1 - K^2} = 0$$

$$\therefore \text{Center} \equiv \left(\frac{5K^2 - 2}{K^2 - 1}, \frac{9K^2 - 3}{K^2 - 1} \right)$$

$$\text{radius} = \frac{3\sqrt{5}}{|K^2 - 1|}$$

for $K = 2$

$$\text{Center} \equiv (6, 11)$$

$$\text{radius} = 2\sqrt{5}$$

$$\text{for } K = \frac{1}{2}$$

$$\text{Center} \equiv (1, 1)$$

$$\text{radius} = 2\sqrt{5}$$

for $K = 3$

$$\text{Center} \equiv \left(\frac{43}{8}, \frac{78}{8} \right)$$

$$\text{radius} = \frac{9\sqrt{5}}{8}$$

$$\text{for } K = \frac{1}{3}$$

$$\text{Center} \equiv \left(\frac{13}{8}, \frac{18}{8} \right)$$

$$\text{radius} = \frac{9\sqrt{5}}{8}$$

2. Ans (C)

$$PA^2 = K^2 \cdot PB^2$$

$$\Rightarrow (x-2)^2 + (y-3)^2 = K^2(x-5)^2 + K^2(y-9)^2$$

$$\Rightarrow x^2(1-K^2) + y^2(1-K^2) - 2x(2-5K^2)$$

$$- 2y(3-9K^2) + (4^2 + 3^2 - 5^2K^2 - 9^2K^2) = 0$$

$$\Rightarrow x^2 + y^2 - 2x \left(\frac{5K^2 - 2}{K^2 - 1} \right) - 2y \left(\frac{9K^2 - 3}{K^2 - 1} \right) + \frac{(4^2 + 3^2 - 5^2K^2 - 9^2K^2)}{1 - K^2} = 0$$

$$\therefore \text{Center} \equiv \left(\frac{5K^2 - 2}{K^2 - 1}, \frac{9K^2 - 3}{K^2 - 1} \right)$$

$$\text{radius} = \frac{3\sqrt{5}}{|K^2 - 1|}$$

$$\text{for } K = 2$$

$$\text{Center} \equiv (6, 11)$$

$$\text{radius} = 2\sqrt{5}$$

$$\text{for } K = \frac{1}{2}$$

$$\text{Center} \equiv (1, 1)$$

$$\text{radius} = 2\sqrt{5}$$

$$\text{for } K = 3$$

$$\text{Center} \equiv \left(\frac{43}{8}, \frac{78}{8} \right)$$

$$\text{radius} = \frac{9\sqrt{5}}{8}$$

$$\text{for } K = \frac{1}{3}$$

$$\text{Center} \equiv \left(\frac{13}{8}, \frac{18}{8} \right)$$

$$\text{radius} = \frac{9\sqrt{5}}{8}$$

3. Ans (D)

$$PA^2 = K^2 \cdot PB^2$$

$$\Rightarrow (x-2)^2 + (y-3)^2 = K^2(x-5)^2 + K^2(y-9)^2$$

$$\Rightarrow x^2(1-K^2) + y^2(1-K^2) - 2x(2-5K^2)$$

$$- 2y(3-9K^2) + (4^2 + 3^2 - 5^2K^2 - 9^2K^2) = 0$$

$$\Rightarrow x^2 + y^2 - 2x \left(\frac{5K^2 - 2}{K^2 - 1} \right) - 2y \left(\frac{9K^2 - 3}{K^2 - 1} \right) + \frac{(4^2 + 3^2 - 5^2K^2 - 9^2K^2)}{1 - K^2} = 0$$

$$\therefore \text{Center} \equiv \left(\frac{5K^2 - 2}{K^2 - 1}, \frac{9K^2 - 3}{K^2 - 1} \right)$$

$$\text{radius} = \frac{3\sqrt{5}}{|K^2 - 1|}$$

$$\text{for } K = 2$$

$$\text{Center} \equiv (6, 11)$$

$$\text{radius} = 2\sqrt{5}$$

$$\text{for } K = \frac{1}{2}$$

$$\text{Center} \equiv (1, 1)$$

$$\text{radius} = 2\sqrt{5}$$

$$\text{for } K = 3$$

$$\text{Center} \equiv \left(\frac{43}{8}, \frac{78}{8} \right)$$

$$\text{radius} = \frac{9\sqrt{5}}{8}$$

$$\text{for } K = \frac{1}{3}$$

$$\text{Center} \equiv \left(\frac{13}{8}, \frac{18}{8} \right)$$

$$\text{radius} = \frac{9\sqrt{5}}{8}$$

4. **Ans (A)**

Let Mid point of AB = P(h, k)

(I) If AB is normal chord.

$$t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_2 + t_1 = -\frac{2}{t_1}$$

$$h = \frac{a(t_1^2 + t_2^2)}{2}; k = a(t_1 + t_2)$$

$$= \frac{a}{2} [(t_1 + t_2)^2 - 2t_1 t_2]$$

$$= \frac{a}{2} \left[\left(\frac{k}{a} \right)^2 - 2t_1 t_2 \right]$$

$$= \frac{k^2}{2a} - a[-2 - t_1^2]$$

$$\left(h - \frac{k^2}{2a} \right) = 2a + at_1^2$$

$$\Rightarrow at_1^2 = \left(h - \frac{k^2}{2a} - 2a \right) = a \left(\frac{2}{t_1 + t_2} \right)^2 = a \left(\frac{2a}{k} \right)^2$$

$$\Rightarrow \left(h - \frac{k^2}{2a} - 2a \right) = \frac{4a^3}{k^2}$$

$$\Rightarrow k^2(2ah - k^2 - 4a^2) = 8a^4$$

$$\Rightarrow k^4 - 2a(h - 2a)k^2 + 8a^4 = 0$$

$$\therefore \text{Locus } y^4 - 2a(x - 2a)y^2 + 8a^4 = 0$$

(II) for focal chord $t_2 = -\frac{1}{t_1} \Rightarrow t_2 t_1 = -1$

$$k = a(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{k}{a}$$

$$h = \frac{a}{2}(t_1^2 + t_2^2) = \frac{a}{2}((t_1 + t_2)^2 - 2t_1 t_2)$$

$$= \frac{a}{2} \left(\left(\frac{k}{a} + 2 \right) \right) = \frac{k^2}{2a} + a$$

$$\Rightarrow k^2 = 2a(h - a)$$

$$\therefore \text{Locus } y^2 = 2a(x - a)$$

(III) If AB subtends 90° at (0, 0)

$$\Rightarrow t_2 = -\frac{4}{t_1} \Rightarrow t_1 t_2 = -4$$

$$\therefore \text{Locus of P} \Rightarrow y^2 = 2a(x + 4a)$$

(IV) If AB is inclined at 45° to axis

$$\Rightarrow \frac{2}{t_1 + t_2} = 1 \Rightarrow t_2 = -t_1 - \frac{1}{2}$$

$$\frac{k}{a} = (t_1 + t_2) = 2$$

$$\therefore \text{Locus } y = 2a$$

5. **Ans (C)**

Let Mid point of AB = P(h, k)

(I) If AB is normal chord.

$$t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_2 + t_1 = -\frac{2}{t_1}$$

$$h = \frac{a(t_1^2 + t_2^2)}{2}; k = a(t_1 + t_2)$$

$$= \frac{a}{2} [(t_1 + t_2)^2 - 2t_1 t_2]$$

$$= \frac{a}{2} \left[\left(\frac{k}{a} \right)^2 - 2t_1 t_2 \right]$$

$$= \frac{k^2}{2a} - a[-2 - t_1^2]$$

$$\left(h - \frac{k^2}{2a} \right) = 2a + at_1^2$$

$$\Rightarrow at_1^2 = \left(h - \frac{k^2}{2a} - 2a \right) = a \left(\frac{2}{t_1 + t_2} \right)^2 = a \left(\frac{2a}{k} \right)^2$$

$$\Rightarrow \left(h - \frac{k^2}{2a} - 2a \right) = \frac{4a^3}{k^2}$$

$$\Rightarrow k^2(2ah - k^2 - 4a^2) = 8a^4$$

$$\Rightarrow k^4 - 2a(h - 2a)k^2 + 8a^4 = 0$$

$$\therefore \text{Locus } y^4 - 2a(x - 2a)y^2 + 8a^4 = 0$$

(II) for focal chord $t_2 = -\frac{1}{t_1} \Rightarrow t_2 t_1 = -1$

$$k = a(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{k}{a}$$

$$h = \frac{a}{2}(t_1^2 + t_2^2) = \frac{a}{2}((t_1 + t_2)^2 - 2t_1 t_2)$$

$$= \frac{a}{2} \left(\left(\frac{k}{a} + 2 \right) \right) = \frac{k^2}{2a} + a$$

$$\Rightarrow k^2 = 2a(h - a)$$

$$\therefore \text{Locus } y^2 = 2a(x - a)$$

(III) If AB subtends 90° at (0, 0)

$$\Rightarrow t_2 = -\frac{4}{t_1} \Rightarrow t_1 t_2 = -4$$

$$\therefore \text{Locus of P} \Rightarrow y^2 = 2a(x + 4a)$$

(IV) If AB is inclined at 45° to axis

$$\Rightarrow \frac{2}{t_1 + t_2} = 1 \Rightarrow t_2 = -t_1 - \frac{1}{2}$$

$$\frac{k}{a} = (t_1 + t_2) = 2$$

$$\therefore \text{Locus } y = 2a$$

6. Ans (D)

Let Mid point of AB = P(h, k)

(I) If AB is normal chord.

$$t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_2 + t_1 = -\frac{2}{t_1}$$

$$h = \frac{a(t_1^2 + t_2^2)}{2}; k = a(t_1 + t_2)$$

$$= \frac{a}{2} [(t_1 + t_2)^2 - 2t_1 t_2]$$

$$= \frac{a}{2} \left[\left(\frac{k}{a} \right)^2 - 2t_1 t_2 \right]$$

$$= \frac{k^2}{2a} - a[-2 - t_1^2]$$

$$\left(h - \frac{k^2}{2a} \right) = 2a + at_1^2$$

$$\Rightarrow at_1^2 = \left(h - \frac{k^2}{2a} - 2a \right) = a \left(\frac{2}{t_1 + t_2} \right)^2 = a \left(\frac{2a}{k} \right)^2$$

$$\Rightarrow \left(h - \frac{k^2}{2a} - 2a \right) = \frac{4a^3}{k^2}$$

$$\Rightarrow k^2(2ah - k^2 - 4a^2) = 8a^4$$

$$\Rightarrow k^4 - 2a(h - 2a)k^2 + 8a^4 = 0$$

$$\therefore \text{Locus } y^4 - 2a(x - 2a)y^2 + 8a^4 = 0$$

$$(II) \text{ for focal chord } t_2 = -\frac{1}{t_1} \Rightarrow t_2 t_1 = -1$$

$$k = a(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{k}{a}$$

$$h = \frac{a}{2}(t_1^2 + t_2^2) = \frac{a}{2}((t_1 + t_2)^2 - 2t_1 t_2)$$

$$= \frac{a}{2} \left(\left(\frac{k}{a} + 2 \right) \right) = \frac{k^2}{2a} + a$$

$$\Rightarrow k^2 = 2a(h - a)$$

$$\therefore \text{Locus } y^2 = 2a(x - a)$$

(III) If AB subtends 90° at (0, 0)

$$\Rightarrow t_2 = -\frac{4}{t_1} \Rightarrow t_1 t_2 = -4$$

$$\therefore \text{Locus of P} \Rightarrow y^2 = 2a(x + 4a)$$

(IV) If AB is inclined at 45° to axis

$$\Rightarrow \frac{2}{t_1 + t_2} = 1 \Rightarrow t_2 = -t_1 - \frac{1}{t_1}$$

$$\frac{k}{a} = (t_1 + t_2) = 2$$

$$\therefore \text{Locus } y = 2a$$

MATHEMATICS

SEC-3

1. Ans (3)

Let BO = 3

$$\alpha = (\theta + \alpha) - \theta$$



ΔAOP & ΔBOP

$$\tan \alpha = \tan((\theta + \alpha) - \theta)$$

$$= \frac{\tan(\theta + \alpha) - \tan \theta}{1 + \tan(\theta + \alpha) \tan \theta}$$

$$= \frac{\frac{15+b}{2} - \frac{b}{2}}{1 + \frac{(15+b)b}{4}} = \frac{2.15}{4 + b^2 + 15b^2}$$

$$\text{similarly } \tan \beta = \frac{3.15}{9 + 15b + b^2}$$

Hence $3 \cot \beta - 2 \cot \alpha$

$$= \frac{9 + 15b + b^2}{15} - \frac{4 + b^2 + 15b^2}{15} = \frac{5}{15}$$

$$\Rightarrow \frac{1}{3 \cot \beta - 2 \cot \alpha} = 3$$

2. Ans (6)

$$f(x, y) = |x - 1| + 2|y - 2| + |xy|$$

$$f(y, x) = |y - 1| + 2|x - 2| + |yx|$$

$$f(x, y) = f(y, x)$$

$$|x - 1| + 2|y - 2| + |xy| = |y - 1| + 2|x - 2| + |yx|$$

$$|x - 1| - 2|x - 2| = |y - 1| - 2|y - 2|$$

Possible case for maximum value of $\alpha + \beta$ is

$$x \geq 2 \text{ \& } 1 \leq y \leq 2 \text{ or } y \geq 2 \text{ and } 1 \leq x \leq 2$$

$$\Rightarrow x - 1 - 2(x - 2) = y - 1 + 2(y - 2)$$

$$-x + 3 = 3y - 5$$

$$x + 3y = 8$$

$$x = 8 - 3y$$

$$\text{When } y = 1, \quad x = 5$$

$$\text{\& } y = 2 \quad x = 2$$

$$\Rightarrow (x + y)_{\max} = 5 + 1 = 6$$

3. **Ans (5)**

$A_1H_{11} = A_2H_{10} = A_3H_9 = A_4H_8 = A_5H_7 = ab$
 $a, G_1, G_2, G_3, G_4, \dots, G_{10}, G_{11}, b$ are in G.P.

$$\Rightarrow G_2G_{10} = G_4G_8 = (G_6)^2 = ab$$

$$\prod_{k=1}^5 (A_k G_{12-2k} H_{12-k})$$

$$= (A_1H_{11}) (A_2H_{10}) (A_3H_9) (A_4H_8) (A_5H_7)$$

$$(G_{10}G_8G_6G_4G_2)$$

$$= (ab)^5 (ab)^2 \sqrt{ab}$$

$$= (ab)^7 \sqrt{ab} = (3^2)^7 \times 3 = 3^{15} = p^q$$

$$\Rightarrow \left[\frac{q}{p} \right] = 5$$

4. **Ans (7)**

$$\frac{3}{x} + \frac{5}{y} = \frac{21}{xy} + 1$$

$$\Rightarrow 3y + 5x = 21 + xy ; (x, y \neq 0)$$

$$\Rightarrow xy - 5x - 3y + 21 = 0$$

$$\Rightarrow x(y - 5) - 3(y - 5) + 6 = 0$$

$$\Rightarrow (x - 3)(y - 5) = -6$$

$$\therefore (x - 3, y - 5) \equiv (1, -6), (-1, 6)$$

$$(2, -3), (-2, 3)$$

$$(3, -2), (-3, 2)$$

$$(6, -1), (-6, 1)$$

$$\therefore (x, y) \equiv (4, -1), (2, 11)$$

$$(5, 2), (1, 8)$$

$$(6, 3), (0, 7) \rightarrow (\text{reject}) (\because x \neq 0)$$

$$(9, 4), (-3, 6)$$

number of possible ordered pairs (x,y) is 7

5. **Ans (8)**

Equation-1:

$$2\sin^2x + 4\sin^2x \cos^2x = 2$$

$$\Rightarrow 2\cos^2x (2\sin^2x - 1) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = \pm \frac{1}{\sqrt{2}} \quad \dots(1)$$

Equation-2 :

$$\frac{2 \tan x}{1 + \tan^2 x} + \frac{1 - \tan^2 x}{1 + \tan^2 x} - \tan x = 0$$

$$\Rightarrow \tan^3 x + \tan^2 x - \tan x - 1 = 0$$

$$\Rightarrow \tan x = \pm 1 \quad \dots(2)$$

$$\text{Also } 2\cos^2 x + \sin x \leq 2$$

$$\text{or } 2\sin^2 x - \sin x \geq 0$$

$$\Rightarrow \sin x \geq \frac{1}{2} \text{ or } \sin x \leq 0 \quad \dots(3)$$

$\Rightarrow$ In $[0, 4\pi]$, number of solutions common to (1), (2) & (3) will be 8.